Sayan Bhowmik

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EDUCATION

2027 — Ph.D., School of Chemical and Biomolecular Engineering, Georgia Institute of Technology, Atlanta, GA, USA (expected)
Dissertation Advisors: Phanish Suryanarayana and Andrew J. Medford

2022 — M.Tech., Department of Chemical Engineering, Indian Institute of Science, Bengaluru, India

2020 — B.E., Department of Chemical Engineering, Jadavpur University, Kolkata, India

PUBLICATIONS

Refereed Journal Articles

2026 — **S. Bhowmik**, A. J. Medford, P. Suryanarayana, "Bulk boundary condition for semi-infinite surfaces in density functional theory," *J. Chem. Theory Comput.*, doi:10.1021/acs.jctc.6c01550.

P. Ray, **S. Bhowmik**, P. Suryanarayana, S. R. Kalidindi, A. J. Medford, "Electronic manifolds for extrapolative alloy discovery," *Digital Discovery* 5, 3850–3862.

2025 — **S. Bhowmik**, A. Sharma, A. J. Medford, J. E. Pask, P. Suryanarayana, "Ab initio study of strain-driven vacancy clustering in aluminum," *Phys. Rev. B* 112, 174105.

S. Bhowmik, A. J. Medford, P. Suryanarayana, "Real-space Hubbard-corrected density functional theory," *J. Chem. Phys.* 163, 234113.

S. Bhowmik, J. E. Pask, A. J. Medford, P. Suryanarayana, "Spectral scheme for atomic structure calculations in density functional theory," *Comput. Phys. Commun.* 308, 109448.

S. Bhowmik, A. Govind Rajan, "Size and chemical environment control nanopore geometry in 2D MoS_2: From irregular to triangular defects," *Small*, 2504611.

2023 — **S. Bhowmik**, J. H. Warner, A. Govind Rajan, "Role of chemical etching in the nucleation of nanopores in 2D MoS_2: Insights from first-principles calculations," *J. Phys. Chem. C* 127, 6873–6883.

2022 **S. Bhowmik**, A. Govind Rajan, "Chemical vapor deposition of 2D materials: A review of modeling, simulation, and machine learning studies," *iScience*, 103832.

2021 **S. Bhowmik**, V. Chakraborty, P. Das, "Batch adsorption of indigo carmine on activated carbon prepared from sawdust: A comparative study and optimization of operating conditions using response surface methodology," *Results in Surfaces and Interfaces* 3, 100011.

Manuscripts in Submission

2026 **S. Bhowmik**, J. E. Pask, P. Suryanarayana, "On-the-fly Δ-machine learned force fields for warm dense matter using a Yukawa reference model: Application to magnesium," submitted to *Physics of Plasmas*, arXiv:2611.00170.

Web-based Publications

2026 A. J. Medford, N.-K. Yu, **S. Bhowmik**, "Replacing exams with 'write your own problem' assignments in an online data analytics course," in D. A. Joyner and N. Roy (Eds.), *Georgia Tech Academic AI Playbook*, 2026 ed., pp. 241–244, Georgia Institute of Technology, hdl.handle.net/1853/81982.

2024 **S. Bhowmik**, "Electronic Structure Theory," 14-lecture open-access video series, Medford Group Training Materials, Georgia Institute of Technology.

AWARDS

2026 CRE Travel Award, Catalysis and Reaction Engineering Division, American Institute of Chemical Engineers

Best Online Teaching Assistant Award, Center for Teaching and Learning, Georgia Institute of Technology

2024 Shell Outstanding Teaching Assistant Award, Georgia Institute of Technology

2023 Best Ph.D. Qualifying Exam Award, Georgia Institute of Technology

Exemplary Academic Achievement Award, Georgia Institute of Technology

2022 Prof. N. R. Kuloor Memorial Award (Gold Medal), Indian Institute of Science

GRANTS AND FELLOWSHIPS

2026 NSF ACCESS Explore Allocation CHM260162 (PI), 200,000 ACCESS Credits, "Bulk boundary conditions for first-principles surface calculations: Validation across metals, transition metals,

semiconductors and insulators, and application to adsorption," Georgia Institute of Technology

Eckert Graduate Fellowship, Georgia Institute of Technology

2020 Graduate Aptitude Test in Engineering (GATE) Fellowship, Ministry of Education, Government of India, Indian Institute of Science

CONFERENCE ACTIVITY

Papers Presented

2026 "Beyond vacuum padding: Bulk-boundary conditions for accurate semi-periodic density functional theory," AIChE Annual Meeting, Minneapolis, MN, USA, November 2026.

"Real-space Hubbard-corrected density functional theory and Hubbard parameters from hybrid functionals," ACS Spring Meeting, Atlanta, GA, USA, March 2026.

"From irregular defects to perfect triangles: Tuning nanopore geometry in 2D MoS_2 via size and chemical environment control," ACS Spring Meeting, Atlanta, GA, USA, March 2026.

2025 "Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum," Society of Engineering Science Annual Technical Meeting, Atlanta, GA, USA, October 2025.

2024 "Ab initio investigation into the aggregation of vacancies and formation of dislocation loops in aluminum," Gordon Research Conference on Computational Materials Science and Engineering, Newry, ME, USA, July 2024. Poster.

"SPARC-atom: An efficient real-space electronic structure calculator for atoms," Gordon Research Seminar on Computational Materials Science and Engineering, Newry, ME, USA, July 2024. Poster.

TEACHING EXPERIENCE

Georgia Institute of Technology, Lecturer, Density Functional Theory Module

Vertically Integrated Projects: Big Data and Quantum Mechanics (Fall and Spring, 2024–present)

Georgia Institute of Technology, Teaching Assistant

Data Analytics for Chemical Engineers (Fall 2025)

Transport Phenomena (Fall 2023)

RESEARCH EXPERIENCE

2022–present Graduate Research Assistant, School of Chemical and Biomolecular Engineering, Georgia Institute of Technology. Advisors: Phanish

Suryanarayana and Andrew J. Medford.
Real-space density functional theory method development in the SPARC electronic structure code, including bulk boundary conditions for surface calculations, Hubbard-corrected density functional theory with a spectral atomic solver, and strain-driven vacancy clustering in aluminum.

2021–2022 Graduate Research Fellow, Department of Chemical Engineering, Indian Institute of Science. Advisor: Ananth Govind Rajan.
Nanopore nucleation and geometry in 2D MoS_2 using density functional theory, Marcus theory, kinetic Monte Carlo, and chemical graph theory.

SERVICE TO PROFESSION

Workshop Instructor, SPARC hands-on training session, 10th i-CoMSE Workshop on QM/MM and Ab Initio Molecular Dynamics, Oklahoma State University, Stillwater, OK, USA, 2025

DEPARTMENTAL AND UNIVERSITY SERVICE

Reviewer, Presidential Undergraduate Research Award, Georgia Institute of Technology, 2024–2025

REFERENCES

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